Explainable Early Warning for Next-Week Abnormal Reported 311 Demand

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Abstract

Urban 311 systems provide large-scale traces of reported municipal demand, but they do not directly measure all underlying service need. We frame NYC 311 modeling as a leakage-controlled early-warning problem: for each neighborhood, feature week, and complaint category, predict whether the next week will exceed its recent local baseline by at least 1.5 standard deviations and at least three requests. The study uses 30.9 million NTA-matched requests to construct a 1.33 million-row modeling panel. The final candidate is a no-shortcut LightGBM model that excludes the 8-week statistics used to construct the target, uses only information available by the end of week t, applies Platt calibration on validation year 2024, and evaluates unchanged on held-out test year 2025. On 2025 test rows, the model obtains PR-AUC 0.3165, F1 0.3613, precision 0.2635, recall 0.5744, precision@1% 0.5697, precision@5% 0.4180, and Brier score 0.0869; cluster-bootstrap 95% intervals are reported for the main metrics. Count-forecasting baselines and sparse-target diagnostics show that the task is substantially harder than simple count extrapolation. SHAP beeswarm and local TP/FP/FN waterfall explanations are computed for the same fitted LightGBM score model, showing that shifted historical demand dominates while service category, current count, calendar, weather, and borough features refine the alert ranking.

Keywords: urban computing, 311 service requests, early warning, reported demand, explainable machine learning, calibration, SHAP

1 Introduction

Urban service agencies increasingly use administrative and resident-generated records to monitor demand and prioritize attention. NYC 311 is a rich example because it records millions of requests across housing, noise, infrastructure, sanitation, water, environmental, and public-safety domains. These records are not a complete census of urban conditions. They mix actual conditions with reporting access, trust, awareness, service expectations, and administrative processing [1, 2]. A forecasting model built on 311 data should therefore be described as forecasting abnormal reported demand, not all underlying need.

We study an operational early-warning task: after feature week t closes, estimate whether a neighborhood and complaint category will experience an abnormal reported-demand event in week $t+1$. The model is intended to rank cases for analyst review, not to automate service allocation. This framing makes three design choices central. First, the target should reduce sparse-panel artifacts, where one new request in a historically zero cell can otherwise become a nominal "surge." Second, the feature matrix should not include variables that directly reconstruct the target threshold. Third, evaluation should emphasize ranking, calibration, uncertainty, workload, and error severity, not only a single F1 score.

The contributions are empirical and methodological rather than a claim of a new algorithm. We provide a reproducible NYC 311 neighborhood-week-category panel, define a minimum-count abnormal reported-demand target, compare count and classification baselines, evaluate rolling-origin and held-out future performance, quantify uncertainty and seed stability, calibrate the score layer, audit workload heterogeneity, and explain the actual fitted score model with global and local SHAP outputs.

2 Related Work

Urban computing connects administrative, geographic, and sensed data to city-scale decision support [3]. Spatiotemporal learning surveys emphasize temporal validation, spatial structure, and careful operational framing in urban prediction problems [4]. Within 311 research, complaint volumes have been used to characterize urban locations [1], but reporting bias can distort data-driven governance decisions [2]. Domain studies connect 311 signals to flooding [5], extreme heat [6], and municipal service design [7]. Our setting differs by evaluating next-week abnormal reported demand across all neighborhoods and complaint categories in a single chronological panel.

The target is rare and imbalanced, so accuracy is not a useful headline metric. Imbalance-learning and evaluation work motivates reporting precision, recall, F1, PR-AUC, ROC-AUC, and precision at operational review budgets [8–11]. Gradient-boosted tree models such as LightGBM are strong tabular baselines for nonlinear interactions [12]. Because alerting systems require transparency, we use TreeSHAP to explain fitted LightGBM score contributions while avoiding causal interpretation of post-hoc explanations [13–15].

3 Methodology

3.1 Panel and Target

Source data are NYC Open Data 311 archives for 2010–2019 and 2020–present [16, 17], 2020 NTA boundaries [18], and NOAA GHCN-Daily observations from Central Park station `GHCND:USW00094728` [19]. The analysis window uses 2015–2025 requests spatially joined to 2020 NTAs and aggregated to Monday-start weeks.

Each row corresponds to one NTA n, one feature week t, and one mapped complaint category c. The nine categories are noise, housing, water/sewer, public safety, parking/traffic, environment, infrastructure, sanitation, and other. The deterministic mapping covers 320 original complaint types. Housing, noise, parking/traffic, sanitation, infrastructure, public safety, water/sewer, environment, and other account for 26.06%, 21.70%, 21.08%, 9.99%, 8.64%, 5.16%, 3.92%, 2.03%, and 1.43% of mapped requests, respectively. The full mapping and 'other' composition are provided as supplementary CSV files. The dense panel has 262 NTAs, 575 weekly periods, and nine categories, or 1,355,850 rows. Relative to this dense panel, 30,654 rows are excluded because of warm-up/history requirements and unavailable next-week targets, leaving 1,325,196 model-ready rows (Table 1).

Table 1 Dataset construction summary.

Item	Value	Item	Value
NTA-matched 311 requests	30,940,129	Match rate	99.9777%
Analytical unit	NTA-week-category	Panel rows	1,355,850
Model-ready rows	1,325,196	Excluded rows	30,654
NTAs	262	Complaint categories	9
Feature weeks	575	NOAA station days	4,018
Final raw predictors	57	Encoded model columns	69

The reference abnormal event is $\text{count}_{n,t+1,c} > \mu^{8w}_{n,t,c} + 1.5\sigma^{8w}_{n,t,c}$, where the mean and standard deviation are computed from a shifted historical eight-week window ending before the target week. To reduce one-call artifacts in near-zero cells, the final target further requires $\text{count}_{n,t+1,c} \geq 3$:

$$y_{n,t,c} = \mathbf{1}\{\text{count}_{n,t+1,c} > \mu^{8w}_{n,t,c} + 1.5\sigma^{8w}_{n,t,c} \ \wedge \ \text{count}_{n,t+1,c} \geq 3\}.$$

The 8-week mean, standard deviation, sum, and ratio used in this target construction are not used as model predictors. On the 2025 held-out test year, this T2 target has 27,277 positives across 247,590 rows in the two-year diagnostic split and reduces the share of positives from cells with $\mu^{8w} < 1$ from 17.96% under the original target to 5.10%.

3.2 Models and Evaluation

The final candidate uses shifted 311 history, current feature-week count, ratios to non-target 12-week history, calendar variables, feature-week weather, service category, and borough identifiers. Weather is a city-level observed exposure for week t; it is not a forecast of week $t+1$. Contemporary built-environment snapshots are not used in the final candidate.

The main classifier is LightGBM with the same no-shortcut feature design across folds. A Platt calibrator is fit only on the validation year and then applied unchanged to the test year. The final-style fold trains through 2023, validates on 2024, and tests on 2025. We also report five expanding-window rolling-origin folds, five stochastic seeds for the final-style fold, and cluster-bootstrap confidence intervals that resample NTA-category clusters. The final-style seed-42 LightGBM fit used for the explainability pass trains in 20.1 seconds on the local workstation; this is a reproducibility diagnostic, not a staffing or deployment latency estimate.

Metrics include PR-AUC, F1, precision, recall, precision@1%, precision@5%, Brier score, log loss, expected calibration error (ECE), and alert rate. Precision@k is reported because agencies can review only a fraction of scored rows. Thresholds are fit on validation only. The resulting alerts should be interpreted as a ranking and triage signal rather than an automatic allocation rule.

4 Results

4.1 Target, Calibration, and Uncertainty

Table 2 summarizes the final-style 2025 evidence for the no-shortcut LightGBM candidate. The Platt-calibrated score improves calibration sharply while preserving the ranking metrics, as expected for a monotone score transformation.

Table 2 Held-out 2025 performance for the final no-shortcut LightGBM candidate. Confidence intervals are cluster-bootstrap 95% intervals over NTA-category clusters.

Metric	Estimate	95% CI or stability
PR-AUC	0.3165	[0.3080, 0.3252]
F1	0.3613	[0.3551, 0.3676]
Precision	0.2635	[0.2582, 0.2690]
Recall	0.5744	[0.5645, 0.5845]
Precision@1%	0.5697	[0.5428, 0.6023]
Precision@5%	0.4180	[0.4035, 0.4324]
Lift@1%	5.1506	[4.8934, 5.4603]
Brier score	0.0869	[0.0852, 0.0888]
Log loss	0.2933	[0.2881, 0.2991]
Five-seed PR-AUC	0.3185	SD 0.0023
Five-seed F1	0.3638	SD 0.0015

Across five rolling-origin folds, the T2 target has mean PR-AUC 0.2997, precision@5% 0.4015, and F1 0.3583. In the final-style 2025 fold, Platt calibration reduces Brier score from 0.1676 to 0.0869 and ECE from 0.2493 to 0.0057. Figure 1 shows the resulting reliability curve.

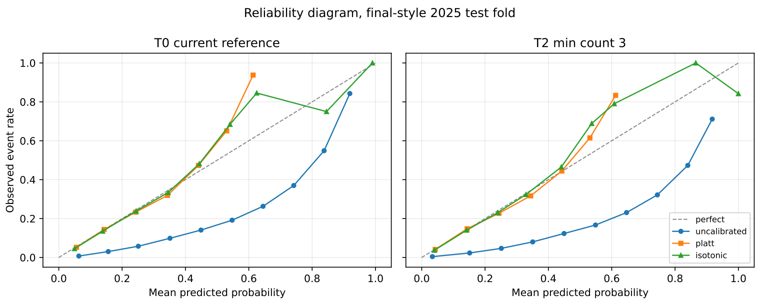

Fig. 1 Reliability diagram for the final-style calibrated LightGBM candidate. Calibration is fit on validation year 2024 and evaluated unchanged on held-out year 2025.

4.2 Baselines and Ablation

Count-model baselines are substantially weaker event rankers than the classification model (Table 3). They remain useful as a sanity check: the classifier is not merely reproducing a count forecast with a threshold.

Table 3 Baseline comparison on the held-out 2025 fold. Formula-threshold count decisions apply the same abnormal-event threshold to predicted counts.

Method	Count MAE	PR-AUC	Precision@5%	F1	Recall
Poisson GLM, no NTA FE	8.6680	0.1400	–	0.1460	–
Poisson GLM + NTA fixed effects	8.6839	0.1400	–	0.1492	–
HGB Poisson count, formula threshold	8.0211	0.1528	0.1755	0.1710	0.1026
Hurdle HGB count, formula threshold	7.9377	0.1517	0.1722	0.1812	0.1108
No-shortcut LightGBM classifier	–	0.3165	0.4180	0.3613	0.5744

Full-row ablations show that history and current count provide most of the signal, while calendar and weather add smaller increments (Table 4). Weather does not improve validation PR-AUC over history plus calendar in this pass, so the weather variables should be read as contextual controls rather than a core contribution. Adding NTA fixed effects slightly improves diagnostic test metrics, but the gain is small enough that final claims should remain conservative until multi-seed evidence is expanded.

Table 4 Full-training ablation for the T2 no-shortcut target. Selection uses validation metrics; 2025 test metrics are diagnostic.

Feature configuration	Val PR-AUC	Val P@5%	Val F1	Test PR-AUC	Test P@5%	Test F1
History lags only	0.1848	0.2352	0.2616	0.1857	0.2381	0.2672
History + current count	0.2464	0.3287	0.3152	0.2528	0.3383	0.3173
History + calendar	0.2754	0.3633	0.3436	0.2861	0.3804	0.3446
History + calendar + weather	0.2738	0.3639	0.3466	0.2955	0.3887	0.3537
Final + borough	0.2939	0.3906	0.3611	0.3165	0.4180	0.3613
Final + NTA fixed effects	0.2951	0.3924	0.3618	0.3187	0.4215	0.3634

4.3 Workload, Groups, and Error Severity

At the validation-selected Platt threshold, the 2025 test year produces a mean of 568.6 alerts per week, with a median of 510 and a range of 97 to 1,055. This volume is high for manual review, so precision@1% and precision@5% are more operationally meaningful than F1 alone. Borough alert rates range from 0.2086 in Manhattan to

0.2592 in Brooklyn; borough precision ranges from 0.2452 in Staten Island to 0.3084 in Manhattan.

Complaint categories differ sharply (Table 5). Noise has the highest alert rate and strongest category F1, while sanitation and other remain weaker. These differences are descriptive performance diagnostics, not a socioeconomic fairness audit. We do not collect new ACS/Census demographic variables in this revision; demographic fairness is therefore a limitation and future-work item rather than a completed result.

Table 5 Held-out 2025 calibrated LightGBM metrics by complaint category.

Category	Positive share	Alert rate	Precision	Recall	F1	PR-AUC
Noise	0.1320	0.3442	0.2915	0.7599	0.4213	0.4127
Water/sewer	0.1091	0.2383	0.2779	0.6070	0.3812	0.3643
Public safety	0.1095	0.2319	0.2824	0.5979	0.3836	0.2996
Housing	0.1216	0.2140	0.2949	0.5193	0.3762	0.3816
Parking/traffic	0.1229	0.2677	0.2605	0.5672	0.3570	0.3035
Infrastructure	0.1236	0.2766	0.2470	0.5529	0.3415	0.2599
Environment	0.0911	0.2392	0.2301	0.6044	0.3333	0.2748
Other	0.0899	0.1532	0.2530	0.4310	0.3188	0.2596
Sanitation	0.0957	0.2050	0.2184	0.4678	0.2978	0.2339

Error-severity analysis shows that the final T2 model recalls 57.44% of positives overall. Recall increases with realized severity: 50.38% for 1.5–2 standard-deviation exceedances, 56.49% for 2–3 standard deviations, and 64.43% for exceedances of at least 3 standard deviations. False negatives therefore still occur in severe events and must be discussed as an operational risk.

4.4 Explainability

SHAP is computed for the same fitted no-shortcut LightGBM score model used in the final-style 2025 candidate. It is not an ensemble proxy and it is not a causal explanation. The beeswarm in Figure 2 shows both direction and dispersion of feature effects. Shifted history dominates the global explanation with 72.59% of summed mean absolute SHAP importance, followed by service category (7.30%), current count (7.20%), feature-week weather (6.30%), calendar (5.54%), and borough (1.07%). The weather share reflects citywide feature-week exposure and seasonal correlation in reported demand, not neighborhood microclimate or a forecast of next week's weather. The highest-ranked individual predictors are `rolling_12w_mean`, `ratio_to_12w_mean`, `rolling_4w_std`, `complaint_count`, and `rolling_12w_std`. The formula-aligned 8-week target-construction fields do not appear as predictors.

Local cases were selected before interpretation: the highest-score true positive, the highest-score false positive, and the missed positive with the largest realized z-exceedance. The TP is a Parkchester/Bronx housing case on 2025-10-20 with score 0.9754 and 244 next-week requests. The FP is a Mapleton-Midwood/Brooklyn noise case on 2025-03-17 with score 0.9563 but no abnormal event. The FN is an Elmhurst/Queens other-category case on 2025-06-30 with score 0.3165 and 426 next-week requests, illustrating that large rare shocks can be missed when recent history and category patterns do not support a high pre-event score.

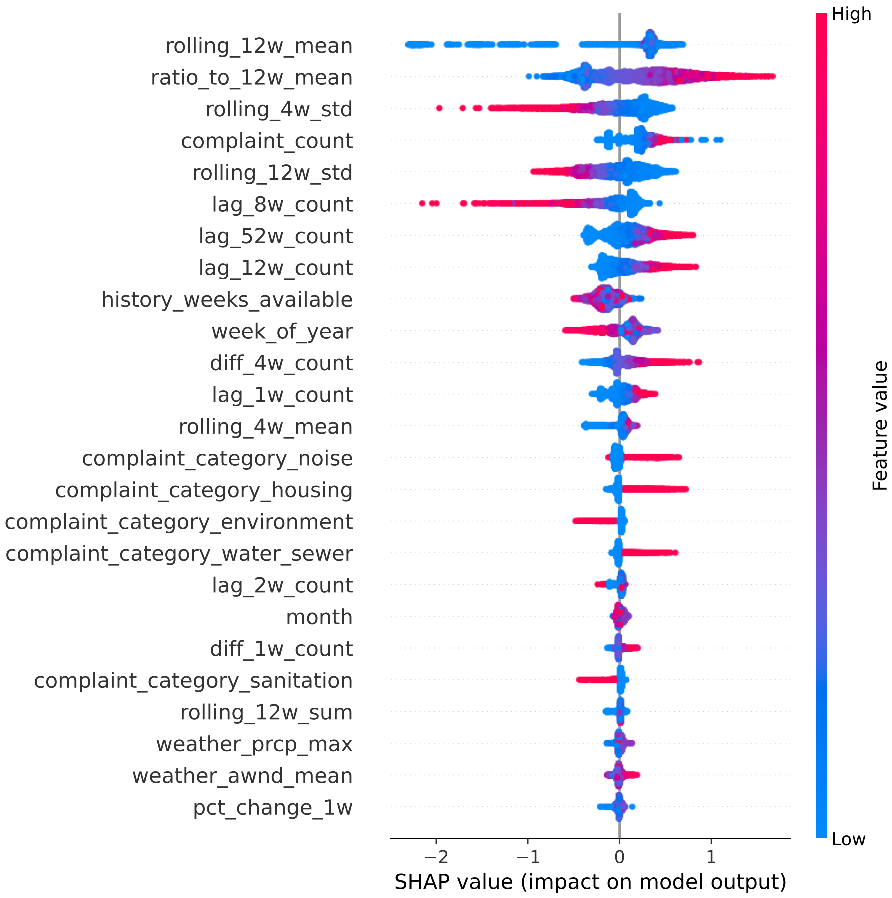

Fig. 2 SHAP beeswarm for the final-style no-shortcut LightGBM score model on a stratified sample of 8,000 held-out 2025 rows. The explanation model and score model are the same fitted LightGBM.

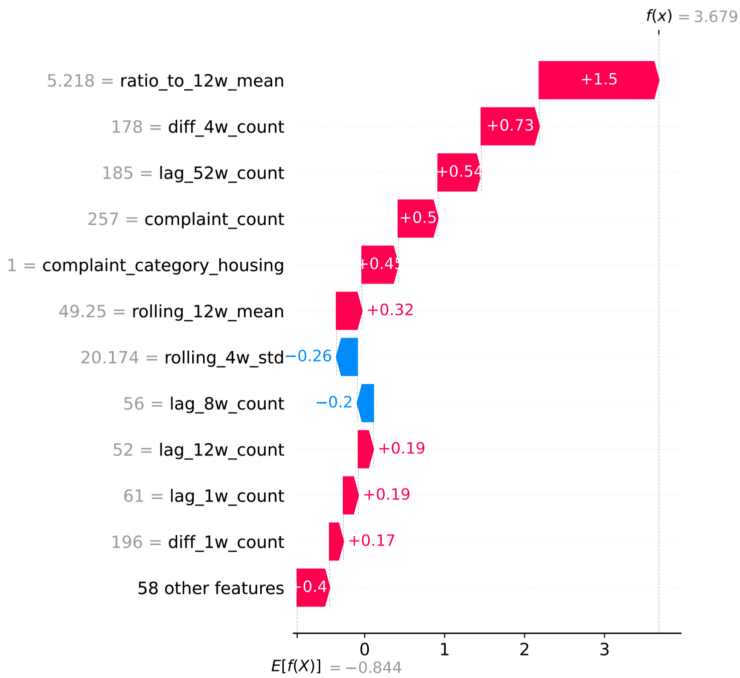

Fig. 3 Local SHAP waterfall explanation for the pre-specified high-confidence true-positive case. The plot explains LightGBM score contributions for the selected row and should not be read as causal attribution.

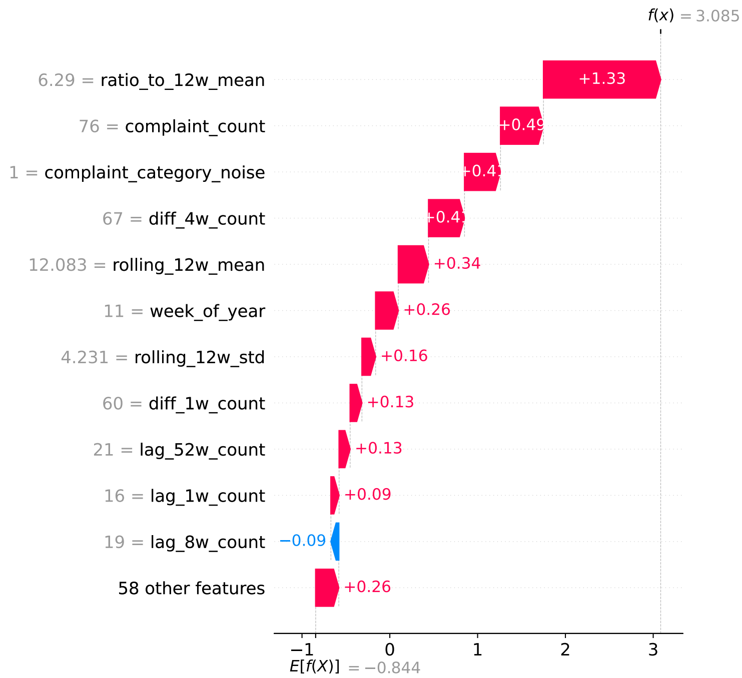

Fig. 4 Local SHAP waterfall explanation for the pre-specified high-confidence false-positive case. The plot explains LightGBM score contributions for the selected row and should not be read as causal attribution.

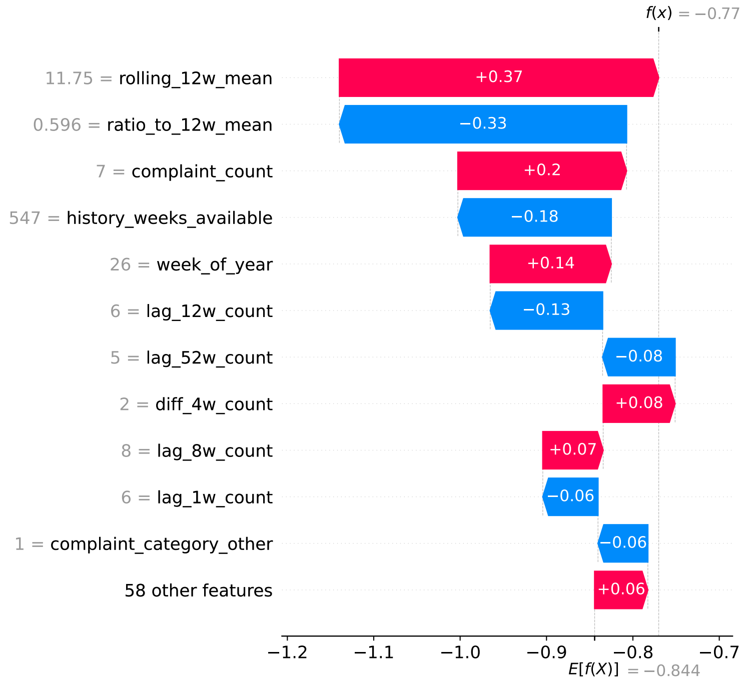

Fig. 5 Local SHAP waterfall explanation for the pre-specified severe false-negative case. The plot explains LightGBM score contributions for the selected row and should not be read as causal attribution.

5 Discussion

The revised evidence supports a narrower claim than the original manuscript. The model can rank next-week abnormal reported-demand risk better than count baselines and simple history-only configurations, but it is best viewed as a triage aid. Precision remains modest at the validation-selected alert threshold, while precision@1% and precision@5% show stronger concentration when analyst capacity is limited. Any deployment could also create feedback: alerts may change agency attention, resident reporting behavior, and future labels, so prospective monitoring after deployment would be required.

The target redesign improves construct validity in sparse panel cells. Under the original threshold, nearly 18% of test positives came from cells with $\mu^{8w} < 1$; the T2 target reduces this to 5.10% by requiring at least three next-week requests. This change also narrows the paper's claim: the task is not raw demand forecasting but early warning for abnormal reported-demand events with a minimum count. A COVID/archive-boundary sensitivity check found visible 2020–2021 prevalence and category-mix shifts; excluding 2020–2021 from training changed 2025 PR-AUC from 0.3165 to 0.3137 and precision@5% from 0.4180 to 0.4122, so the retained training window is treated as a documented sensitivity rather than a homogeneous time series.

Several limitations remain. 311 data reflect reported demand and can reproduce spatial and social reporting inequities. Weather is city-level observed exposure for week t, not a spatial forecast for week $t + 1$. Historical source vintages are unavailable, the modeling panel does not retain row-level archive-source identifiers, and weekly counts may be affected by late entry, reclassification, deduplication, or open-data revision. We therefore cannot fully measure production-time data delay or archive-boundary provenance from the current panel. We do not collect new demographic, staffing, cost, or operational outcome data in this revision, so socioeconomic fairness, cost-sensitive thresholds, and downstream service impact remain future work. The current NTA fixed-effect gain is small and should be treated as suggestive rather than definitive.

6 Conclusion

We present a leakage-controlled early-warning study for next-week abnormal reported NYC 311 demand. The final candidate uses a minimum-count target, removes target-construction shortcut predictors, evaluates chronologically, calibrates probabilities on validation only, and reports uncertainty, seed stability, workload, and local explanations. On held-out 2025 rows, the Platt-calibrated no-shortcut LightGBM model reaches PR-AUC 0.3165, F1 0.3613, recall 0.5744, precision@1% 0.5697, and precision@5% 0.4180. The strongest practical use is not automatic dispatch, but capacity-aware prioritization of analyst attention toward reported-demand risks that merit review.

Declarations

Data and code availability. Source data include NYC Open Data 311 archives `76ig-c548` and `erm2-nwe9`, 2020 NTA boundaries `9nt8-h7nd`, and NOAA GHCN-Daily observations from Central Park station `GHCND:USW00094728`. Scripts, complaint-type mappings, configurations, evaluation code, and compact reviewer-facing outputs are available at `https://github.com/PhongLam26/SIMC_NYC`. Large intermediate panels can be rebuilt from the public sources using the repository scripts.

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Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le served as academic advisor and supervised the study design, methodology, and manuscript editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed validation and review.

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